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Homework for Chap 11

Which trend would you expect for the zone boundary LA and TA phonons when going from ZnO via ZnS and ZnSe to ZnTe. Compare your thought with data in the literature. What do you expect for zone center optic phonons?

① 对声学声子: 振动频率 $\omega \propto \sqrt{\frac{k}{\mu}}$ (弹簧振子模型)
 k : 弹性常数, μ 约化质量

$M_O < M_S < M_{Se} < M_{Te}$, 因此

对 LA, TA: $\omega_{ZnO} > \omega_{ZnS} > \omega_{ZnSe} > \omega_{ZnTe}$

查数据

	ZnO	ZnS	ZnSe	ZnTe
LA (THz)	10	3.7	3.09	2.46
TA (THz)	4.92	2.03	1.59	1.23

与预测相同

② 对光学声子

$ZnO \rightarrow ZnTe$ 约化质量上升 (振动质量上升) TO, LO 下降

离子性下降, 极性减弱 $\Delta(LO-TO)$ 下降

对 LO, TO: $\omega_{ZnO} > \omega_{ZnS} > \omega_{ZnSe} > \omega_{ZnTe}$

$\Delta_{ZnO} > \Delta_{ZnS} > \Delta_{ZnSe} > \Delta_{ZnTe}$